

RUNNING A CLOUD ON ONE BOX

A bootstrapped Function-as-a-Service platform on Firecracker microVMs and Hetzner dedicated metal

Internal founding whitepaper - strategy, architecture, unit economics, sensitivity and risk

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All forward-looking figures are planning estimates derived from the model of record. They are not commitments, valuations or investment advice. Update this document whenever the spreadsheet assumptions change; where the two disagree, the spreadsheet wins.

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1. Executive summary

This document is the founding reference for a single-operator cloud platform: a Function-as-a-Service (FaaS) product built on Firecracker microVMs, hosted on one Hetzner EX44 dedicated server (€44/month rent, €64.25/month all-in including amortised setup, backups, object storage, DNS and monitoring). It consolidates the financial model of record (ex44_faas_financial_model.xlsx), the architecture it implies, a full register of assumptions, sensitivity analysis, the competitive picture as of July 2026, and the risk and mitigation plan.

The core arithmetic. One EX44 carries 64 GB of RAM. After reserving 8 GB for the host OS and the co-located control plane, and holding 15% headroom for traffic spikes, 47.6 GB is sellable tenant RAM. Over a 730-hour month that is 34,748 sellable GB-RAM-hours, so a fully loaded GB-RAM-hour costs €0.00185 to produce. AWS Lambda retails the equivalent unit at roughly \$0.06 - about thirty times our production cost - and mid-market platforms (Fly.io, Railway) sit at four to seven times it. That umbrella is the entire business.

The unit economics. Four plans (Free, Hobby €9, Pro €29, Scale €99) monetise concurrent sandbox capacity plus a metered overage at €0.01 per GB-RAM-hour. At the planned customer mix the blended ARPU is €21.60, the physical cost to serve a paying customer - including the four free-tier users each one statistically drags along - is €0.44, and Stripe takes €0.93. Contribution margin is €20.23 per customer per month (93.7%). Fixed costs are cleared by **4 paying customers**; one full box yields €3,520.80 of monthly revenue against €215.25 of all-in cost, a 93.9% margin.

The trajectory. In the base growth scenario (2 new customers in month 1, adds compounding 30% monthly, 5% monthly churn) the platform is profitable from month 2, cash-positive cumulatively from month 3, and exits month 12 with 130 paying customers, ~520 free users, €2,808 MRR and €2,623 monthly profit, having consumed roughly 60% of one box's resident capacity. Total capital at risk before break-even: about €62.

What could kill it. The model is margin-robust but capacity-fragile. Four things matter: (1) snapshot-size drift - at a 300 MB average instead of the planned 130 MB, the per-box ceiling collapses from 163 to 70 customers; (2) the build pipeline sharing RAM with production microVMs - an unbounded build can OOM the box; (3) single-server hardware failure before a failover pair exists - mitigated by a second box around month 6, when MRR covers it roughly eightfold; (4) churn above ~8%/month, which makes paid acquisition untenable (LTV falls from €405 to €253). Every one of these has a cheap, known mitigation described in section 11.

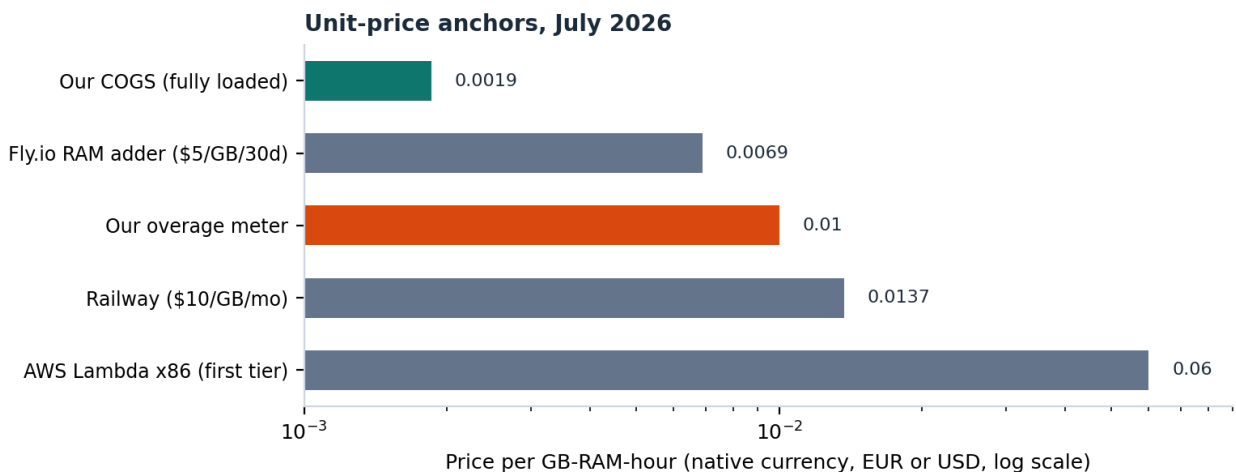
Verdict: green light. The plan turns €64/month of infrastructure into a €2,800/month business inside a year with no external capital, and each additional box is self-funding. The moat is not technology - Firecracker is open source - it is cost structure, focus and operational discipline. This document exists to protect that discipline.

2. Thesis and market context

2.1 The pricing umbrella

Serverless compute is priced, industry-wide, in memory-time. AWS Lambda charges \$0.0000166667 per GB-second on x86 in its first pricing tier - \$0.06 per GB-hour - plus \$0.20 per million requests. Railway meters at \$10 per GB of RAM per month (\$0.0137/GB-hour). Fly.io prices additional RAM at roughly \$5 per GB per 30 days (\$0.007/GB-hour for the RAM component alone; effective machine pricing including CPU lands around \$0.008-0.012). Render sells fixed always-on instances from about \$7/month. Against these anchors, our fully loaded production cost on Hetzner metal is €0.00185 per GB-hour. The market's retail floor sits 4x above our cost; its ceiling (Lambda) sits about 30x above it.

The gap persists for structural reasons that do not apply to us: hyperscalers carry enormous margins, egress fees and enterprise sales costs; the mid-market PaaS layer mostly resells hyperscaler capacity or runs venture-funded operations with payrolls to feed. A solo operator on owned-rented metal with near-zero payroll keeps almost the entire spread.



Native-currency comparison; EUR/USD are close enough to parity that the ordering is unaffected. Cloudflare Workers is omitted from the chart because it bills CPU-milliseconds on V8 isolates rather than resident RAM-time - see section 10.

2.2 Why one person can now run a cloud

Two enabling shifts converged. First, **Firecracker** - the virtual machine monitor AWS built for Lambda and Fargate and open-sourced - provides hardware-virtualised (KVM) isolation with less than 5 MiB of overhead per microVM, ~125 ms boot times, and the ability to create up to 150 microVMs per second per host. This gives a single operator what previously required a virtualisation team: the ability to run *untrusted, arbitrary customer code* at container-like density with VM-grade security boundaries. The model budgets a conservative 8 MB per VM to cover the VMM plus jailer, TAP networking and host-side slack.

Second, **commodity dedicated metal** has become absurdly cheap relative to cloud. The Hetzner EX44 - Intel Core i5-13500 (6P+8E, 20 threads), 64 GB DDR4, 2x512 GB NVMe Gen4 in RAID 1, unmetered 1 Gbit uplink - lists at about €39.44/month as of early 2026 (the model conservatively carries €44 plus a €39 setup fee amortised over twelve months). Hetzner adjusted prices again in June 2026, so the exact quote should be re-checked at ordering time; the sensitivity analysis in section 9.9 shows the business is indifferent across the €39-50 range.

2.3 The wedge: idle-heavy applications

The economics rest on a behavioural fact: most deployed applications are idle almost all the time. A hobbyist's webhook receiver, a side project's API, an agency's staging environment - these need

to exist continuously but run rarely. Static concurrency (a VPS, a Render instance) charges for 24/7 residence. Our scale-to-zero design parks idle applications as ~130 MB snapshots on NVMe and restores them in a fraction of a second on the first request, so the marginal cost of a parked app is disk, not RAM. The pricing tabs model this as *average actual concurrency*: a Hobby user whose plan caps 2 concurrent sandboxes keeps only ~0.15 resident on average. That factor - not the cap - is what the box must physically carry, and it is what makes 163 customers per box possible where naive static sizing would allow perhaps two dozen.

2.4 Who we are not competing with

We are not competing with AWS for enterprises, nor with Cloudflare for global edge latency. The honest competitive set is: a €5 always-on VPS the developer must babysit; Fly.io and Railway at 4-7x our unit price; and free tiers that expire or throttle. Secondary angles that cost nothing to hold: EU data residency (Hetzner Germany/Finland) and flat, predictable pricing - a real differentiator against usage-billing anxiety. The strategy is to be the obviously sane place to park many small, idle-heavy, full-runtime workloads.

3. Product definition

3.1 The value metric: GB-RAM-hours

Everything is denominated in one unit: the **GB-RAM-hour** - one gigabyte of guest memory resident for one hour. It is the correct metric for three reasons. It tracks the box's true scarce resource, so price maps to cost by construction. It is industry-legible: Lambda bills GB-seconds, Fly bills RAM-time, so buyers can comparison-shop in our favour. And it is honest to customers: idle costs nothing, because idle applications are snapshotted off RAM entirely.

Plans bundle three entitlements: **deployed sandboxes** (apps parked on disk), a **concurrency cap** (simultaneously running microVMs), and an **included compute quota** in GB-RAM-hours. Usage beyond quota meters at €0.01 per GB-RAM-hour - roughly 5.4x our production cost, ~6x cheaper than Lambda, and adjacent to Fly's effective rate.

3.2 Plan architecture

	Free	Hobby	Pro	Scale
Price (€/mo)	0	9	29	99
Deployed sandboxes (parked)	1	5	25	100
Max concurrent sandboxes	1	2	5	20
Max RAM per sandbox (MB)	128	256	512	1,024
Included compute (GB-h/mo)	5	50	250	1,500
Expected avg resident concurrency	0.02	0.15	0.60	3.00
Share of paying customers	-	55%	40%	5%

The last two rows are planning assumptions, not customer-visible promises: expected resident concurrency drives capacity and COGS; the mix drives blended ARPU.

3.3 How the money mechanics work

A customer subscribes to a plan through Stripe; the subscription is the bundle. Every running microVM is metered per second as guest RAM x runtime. Usage within quota costs nothing extra; usage beyond it appears as overage at €0.01/GB-h on the monthly invoice. Plans are priced so the bundle sells its quota at roughly 3-8x COGS and the meter sells marginal units at ~5x COGS. The model assumes a modest 10 GB-h of average billable overage per paying customer (€0.10/month) - deliberately negligible, so the P&L never depends on overage materialising.

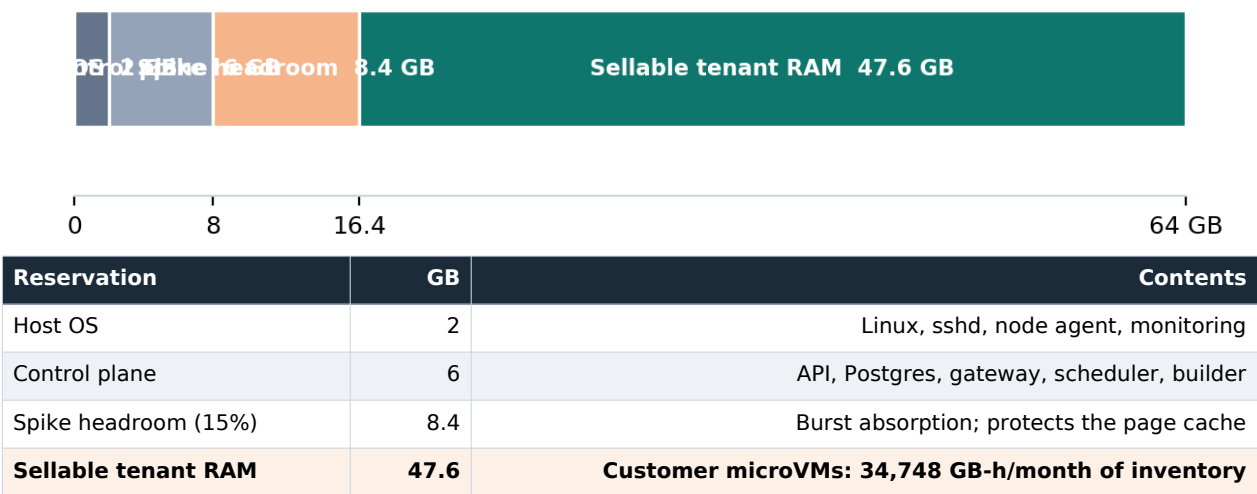
The free tier is a marketing expense, precisely priced. A free user's expected cost is €0.0041/month (compute €0.0036 at 0.02 resident concurrency of a 136 MB VM, plus €0.0005 of snapshot disk). A thousand free users cost €4.09/month - less than the backup line item. The binding cost of the free tier is not euros but *disk slots*: each paying customer's four statistical free riders park four snapshots, and section 9.7 shows the customer ceiling moves from 180 to 138 as the free ratio runs from 2 to 8. Free-tier discipline is a capacity policy, not a cost policy.

4. System architecture

4.1 One-box topology

Everything runs on the EX44. The edge is a reverse proxy/API gateway terminating TLS and routing per-app subdomains. The control plane - REST API, Postgres (state, metering events, billing), the scheduler, and the build service - shares the box inside a fixed 6 GB reservation. The data plane is Firecracker microVMs launched via the jailer with per-VM TAP interfaces and cgroup CPU quotas. Storage splits three ways: the RAID-1 NVMe pair holds the OS, base images and the snapshot store; an external object-storage bucket (€5/month) holds build artifacts and cold data; a Hetzner Storage Box (€4/month) receives off-host backups (Postgres WAL plus periodic snapshot sync).

Where the 64 GB goes



4.2 The sandbox lifecycle

Deploy: customer pushes code; the builder produces a rootfs (buildpacks or Dockerfile). Park: the app's memory-plus-rootfs state is captured as a compressed snapshot, ~130 MB on average, and the microVM is destroyed - the app now costs disk only. Wake: a request hits the gateway; the scheduler restores the snapshot into a fresh jailed microVM - Firecracker restores are of the same order as its ~125 ms cold boot, so first-request latency stays sub-second. Serve, then idle-timeout back to parked. This loop is the product; everything else is bookkeeping around it.

4.3 Scheduling and resource governance

Admission control enforces per-plan concurrency caps before a VM starts. CPU is overcommitted 8:1 (20 threads → 160 vCPU slots), tolerable because the workload is idle-heavy; cgroups cap per-VM CPU shares so one hot tenant degrades gracefully rather than seizing the box. RAM is never overcommitted beyond the 85% utilisation target - the scheduler refuses wakes past the headroom line and sheds oldest-idle VMs first. Per-VM budget includes the 8 MB VMM overhead.

4.4 The build subsystem - the known hard part

Builds (npm install, pip compiling wheels) are the only workload on the box that is both memory-hungry and unpredictable, and they run on the same metal as production microVMs. Policy, from day one: a global build semaphore (at most 2 concurrent builds), a hard memory cgroup per build (~2 GB inside the control-plane reservation), deployment archive capped at 100 MB on Free and Hobby, build timeout with kill-and-requeue, and queueing rather than parallelism

under pressure. If build queue latency ever becomes a sales objection, the correct fix is a €5-10 cloud VM as a remote builder - not loosening the semaphore. Risk R1 tracks this.

4.5 What is deliberately not built

No Kubernetes, no multi-region mesh, no autoscaling groups, no service catalogue. One box, one binary-ish control plane, Postgres, Firecracker, and rsync-class tooling. Boring technology is a load-bearing decision: the operator is also the support desk, and every moving part is a pager duty. Complexity is added only when a paying constraint demands it (see the roadmap gates in section 12).

5. Assumption register

Every input of the model of record, its value, its rationale, and a fragility grade: how badly the business bends if the assumption is materially wrong. Grades: **A** - robust, verified or contractual; **B** - reasonable estimate, moderate consequences; **C** - fragile, actively monitor. The C-grade rows are exactly the ones instrumented in section 9.

Assumption	Value	Rationale / source	Grade
EX44 monthly rent	€44	List ~€39.44 (Feb 2026); carried conservatively. Re-quote at order.	A
Setup fee, amortised 12 mo	€39	Hetzner setup €39-99 by region; promos often waive it.	A
Total RAM	64 GB	EX44 spec, DDR4.	A
Usable disk (RAID 1)	512 GB	2x512 GB NVMe Gen4, mirrored.	A
CPU threads	20	i5-13500: 6P+8E cores.	A
Host OS reserve	2 GB	Linux, sshd, agents.	A
Control plane reserve	6 GB	API, Postgres, gateway, builder co-located.	B
Firecracker overhead / VM	8 MB	Spec is <5 MiB; padded for jailer + TAP slack.	A
RAM utilisation target	85%	Spike headroom; protects page cache.	B
Disk reserve (OS, images, logs)	60 GB	Host OS, kernels, base rootfs layers.	B
Avg snapshot size	130 MB	Compressed memory + diff rootfs. Most fragile number in the model.	C
CPU overcommit	8x	Idle-heavy FaaS tolerates 6-10x.	B
Hours / month	730	365x24/12.	A
Backup (Storage Box)	€4	~€3.9 for 1 TB off-host.	A
Domain / DNS / email	€3	Amortised domain + cheap transactional tier.	A
Object storage / registry	€5	Artifacts off-box; 0 if kept on EX44.	A
Monitoring / misc buffer	€5	Grafana Cloud free tier + surprises.	B
Founder salary	€0	Ramen mode; section 9.8 prices real draws.	B
Stripe fee	2.9% + €0.30	Conservative non-EU card rate; EU entity would pay ~1.5% + €0.25.	B
Monthly churn (paying)	5%	Typical early self-serve; C because LTV/CAC hangs on it.	C
Free users per paying	4	Free tier rides along; binds disk slots, not euros.	B
Overage per customer	10 GB-h/mo	Deliberately negligible (€0.10); upside not load-bearing.	A
Avg resident concurrency	0.02/0.15/0.60/3.0	Free/Hobby/Pro/Scale. The scale-to-zero bet itself.	C
Customer mix	55/40/5	Hobby/Pro/Scale share of paying customers.	B
Growth scenarios	1+20% / 2+30% / 4+40%	New adds in M1, monthly compounding of adds.	C

Reading of the register: the business stands on three C-grade numbers - snapshot size, resident concurrency, and churn. All three are cheap to instrument from week one (p95 snapshot MB, resident GB per customer, cohort retention). The monthly ritual is to replace these estimates with telemetry and re-run the sheet.

6. Capacity model: what one EX44 actually holds

6.1 Three constraint paths

RAM path. 64 GB minus 8 GB of reservations leaves a 56 GB tenant budget; at the 85% utilisation target, 47.6 GB is spendable. The model blends the average resident microVM at 404.8 MB (plan-share-weighted guest RAM of 396.8 MB plus 8 MB overhead), giving **120 safe resident microVMs**. **CPU path.** 20 threads x 8 overcommit = 160 vCPU slots - not binding. **Disk path.** 512 GB minus a 60 GB reserve leaves 452 GB of snapshot budget; at 130 MB average that is **3,560 parked sandboxes**.

Demand side: a paying customer brings 0.5525 expected resident VMs (own plan usage plus four free riders at 0.02 each) and 21.75 parked snapshots (own deployed slots plus four free-tier singles). Dividing supply by demand: concurrency supports 217 customers; disk supports 163. **The model's ceiling is 163 paying customers per box, disk-bound** - at which point the box bills €3,520.80/month.

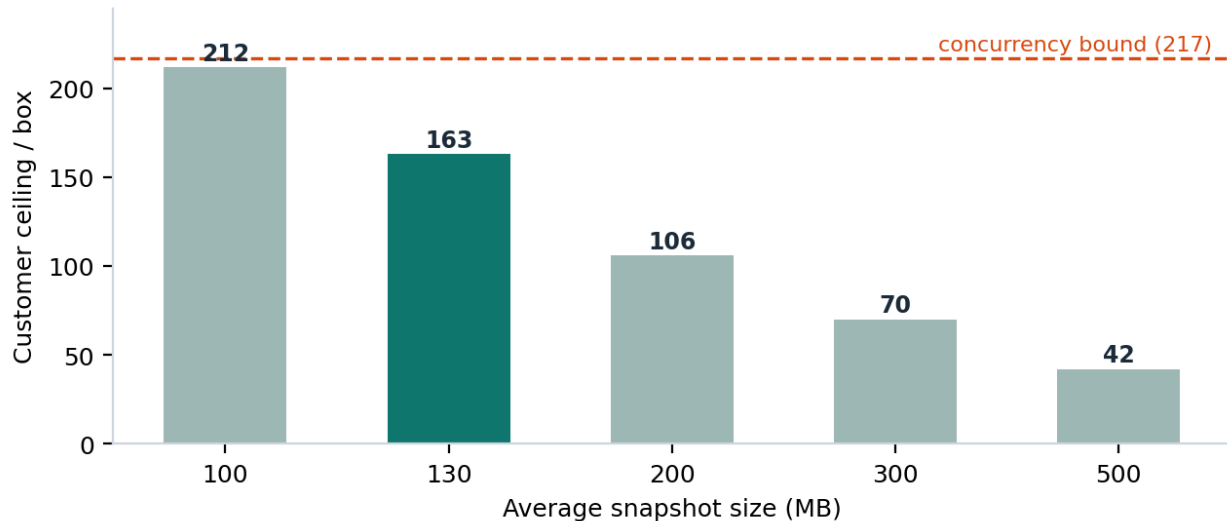
Constraint	Supply	Per-customer demand	Ceiling
RAM (resident VMs)	120 VMs (47.6 GB / 404.8 MB)	0.5525 resident VMs	217
CPU (vCPU slots)	160 slots (20 thr x 8)	-	not binding
Disk (parked snapshots)	3,560 snaps (452 GB / 130 MB)	21.75 snapshots	163 ◀ binding

6.2 Model integrity note: the two RAM blends (recommended fix)

Cross-checking the sheets surfaces one internal inconsistency worth fixing. The Capacity tab sizes the average resident VM by *plan share* (55/40/5 → 404.8 MB), but plans with more RAM also run *more concurrent VMs*, so the demand-weighted average is larger: weighting by expected concurrency gives 312.3 MB of resident RAM per paying customer, i.e. ~565 MB per average resident VM. Under that stricter blend the RAM path supports 47.6 GB / 312.3 MB ≈ **156 customers**, slightly below the disk ceiling of 163 - the binding constraint flips from disk to RAM.

Financially this is noise (156 x €21.60 = €3,370 vs €3,520 at ceiling); operationally it matters, because it says RAM pressure will be felt *before* disk pressure, and month-12 utilisation in the base case is ~83% of the spike-headroom target on the demand-weighted basis versus the 60% the sheet reports on a VM-count basis. Recommendation: change Capacity!B5 to the concurrency-weighted blend, plan the second box against a ~156-customer practical ceiling, and treat 163 as the disk bound it is.

The ceiling is a hostage of snapshot size



Sensitivity of the per-box customer ceiling to average snapshot size (sheet logic, disk path). The 130 MB plan assumption is the teal bar; section 9.2 gives the table.

6.3 Utilisation at the end of year one

At month 12 of the base case (130 paying, 520 free): ~72 resident VMs against 120 slots (60%), ~2,830 parked snapshots against 3,560 (79%), and 83% of the RAM spike-headroom target on the demand-weighted blend. All three gauges agree: one box comfortably carries year one, and the second box is a growth decision (and an availability decision - section 11, R3), not a capacity emergency.

7. Cost structure and unit economics

7.1 Fixed monthly costs

Line item	€/month
EX44 rent	44.00
Setup fee amortised (12 mo)	3.25
Backup Storage Box	4.00
Domain / DNS / email	3.00
Object storage / registry	5.00
Monitoring / misc	5.00
Total fixed (one box)	64.25

Bandwidth is a structural zero: the EX44 uplink is unmetered 1 Gbit. There is no egress line item to manage - a quiet but real advantage over every hyperscaler-hosted competitor.

7.2 The number the business stands on

€64.25 of monthly cost divided by 34,748 sellable GB-RAM-hours = **€0.0018490 per GB-RAM-hour, fully loaded**. Every price in the catalogue is a multiple of this number: the overage meter at €0.01 carries an 81.5% gross margin; the plan bundles sell their quotas at 3-8x it. Because the denominator is fixed capacity, COGS per unit falls as nothing - it is already fully loaded at zero customers, which is why break-even arrives at four.

7.3 Per-plan economics (per user per month)

	Free	Hobby	Pro	Scale
Resident GB-RAM-hours	1.94	28.23	222.42	2,207.11
Cost: compute (€)	0.0036	0.0522	0.4113	4.0810
Cost: snapshot disk (€)	0.0005	0.0025	0.0127	0.0508
Total cost to serve (€)	0.0041	0.0547	0.4240	4.1318
Revenue (€)	0	9.00	29.00	99.00
Gross profit (€)	-0.0041	8.95	28.58	94.87
Gross margin	-	99.4%	98.5%	95.8%

7.4 Blended per-customer economics

Component	€/customer/month
Blended subscription ARPU (55/40/5 mix)	21.50
Overage revenue (10 GB-h avg)	0.10
Total ARPU	21.60
Cost to serve (own plan + 4 free riders + overage COGS)	-0.44
Stripe (2.9% + €0.30)	-0.93
Contribution margin	20.23 (93.7%)

The payment processor costs twice the datacenter. Stripe's €0.93 per customer per month is 2.1x the €0.44 physical cost to serve. Two consequences: an EU entity (fees ~1.5% + €0.25) is worth €0.35/customer/month forever (section 9.5), and annual billing - twelve fixed fees collapsing to one - saves another €3.30 per customer-year, roughly 1.3% of ARPU, before considering its churn-suppressing effect.

7.5 Lifetime value, acquisition budget, break-even

Metric	Value	Derivation
Expected customer lifetime	20 months	1 / 5% monthly churn
LTV (contribution basis)	€404.65	€20.23 x 20
Max sane CAC (LTV:CAC = 3)	€134.88	Ads, content, or founder time per acquired customer
Break-even	4 paying customers	€134.25 / €20.23, rounded up; 2.5% of one box's ceiling
One full box: revenue	€3,520.80 /mo	163 x €21.60
One full box: all-in costs	€215.25 /mo	Fixed + Stripe (usage COGS embedded in box rent)
One full box: profit	€3,305.55 /mo	93.9% net margin at ceiling

Accounting note, from the model of record: the projection charges the full box rent as the fixed cost and reports usage COGS separately as capacity information, so compute cost is never double-counted.

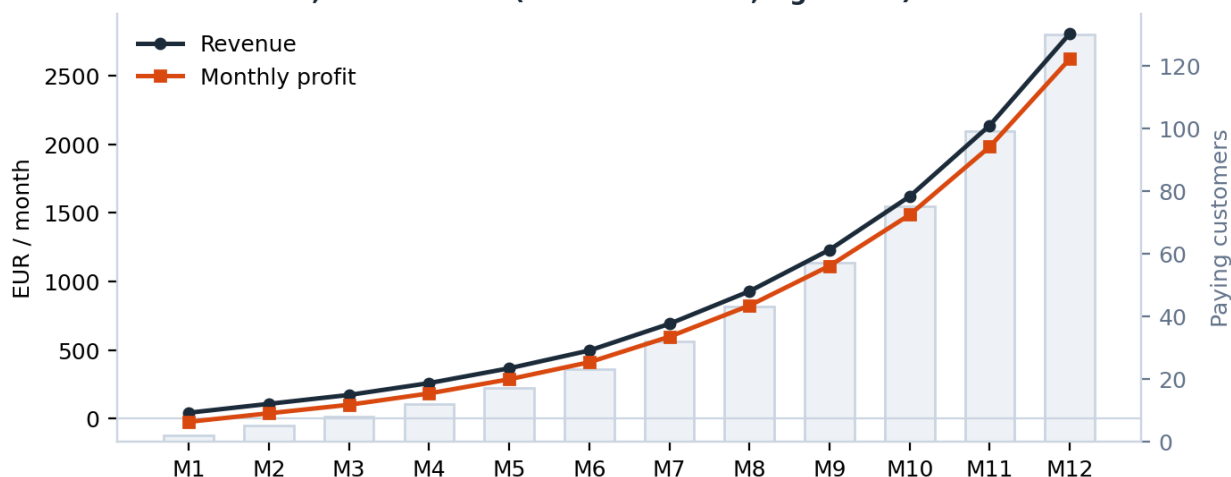
8. Twelve-month projection

8.1 Base scenario, month by month

Base scenario: 2 new paying customers in month 1, new adds compounding at 30% monthly, 5% monthly churn on the existing base, mix and ARPU as modelled. Month-1 cash includes the €39 setup fee.

	New	Churn	Paying	Free	Revenue €	Stripe €	Profit €	Cum. cash €	Cap.
M1	2	0	2	8	43.2	1.9	-22.9	-61.9	1%
M2	3	0	5	20	108.0	4.6	39.1	-22.8	2%
M3	3	0	8	32	172.8	7.4	101.1	78.4	4%
M4	4	0	12	48	259.2	11.1	183.8	262.2	6%
M5	6	1	17	68	367.2	15.7	287.2	549.4	8%
M6	7	1	23	92	496.8	21.3	411.2	960.6	11%
M7	10	1	32	128	691.2	29.6	597.3	1,557.9	15%
M8	13	2	43	172	928.8	39.8	824.7	2,382.7	20%
M9	16	2	57	228	1,231.2	52.8	1,114.1	3,496.8	26%
M10	21	3	75	300	1,620.0	69.5	1,486.3	4,983.1	35%
M11	28	4	99	396	2,138.4	91.7	1,982.4	6,965.5	46%
M12	36	5	130	520	2,808.0	120.4	2,623.3	9,588.8	60%

Base scenario, months 1-12 (bars: customers, right axis)

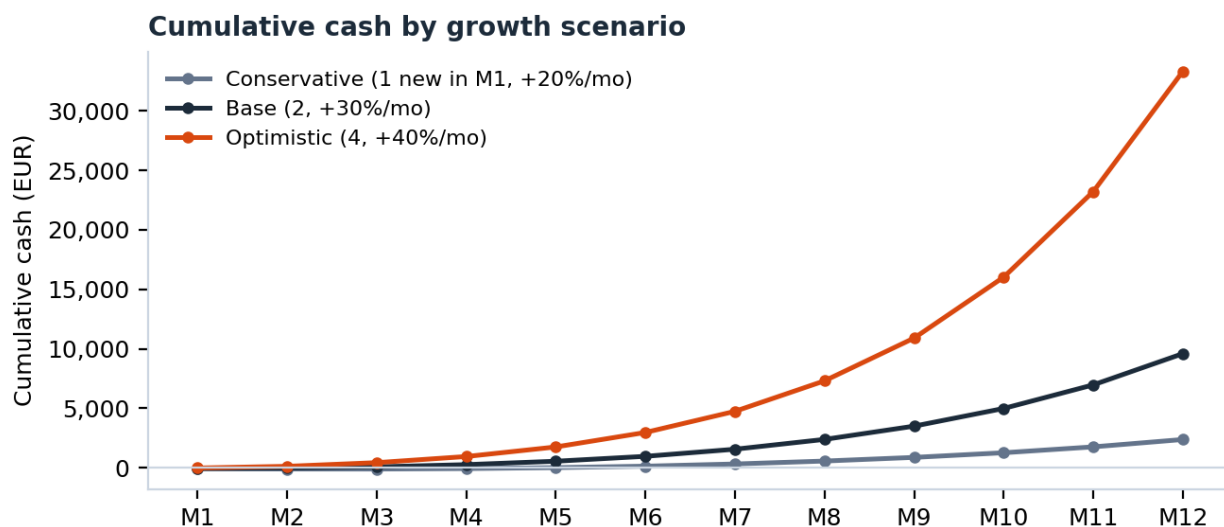


Milestones: first profitable month M2; cumulative cash positive M3 (total capital at risk €62); covers a €500 founder draw from M7 and €1,000 from M9; exits M12 at €2,808 MRR with 60% of resident capacity used (83% on the demand-weighted RAM basis of section 6.2). Churn discipline note: at 130 customers, 5% churn means finding 7 replacement customers a month before any net growth - retention work compounds exactly as acquisition does.

8.2 Scenario comparison

Scenario	M12 paying	M12 MRR €	M12 profit €	Cum. cash €	Profitable	Cash+
Conservative (1, +20%)	33	712.80	618.00	2,373.70	M4	M5
Base (2, +30%)	130	2,808.00	2,623.30	9,588.80	M2	M3
Optimistic (4, +40%)	501	10,821.60	10,100.50	33,306.50	M1	M2

The optimistic path crosses one box's ceiling around M9-10 and ends M12 needing four boxes (501 customers); its profit figure includes the extra rents. Even the conservative path - one customer in month one, slow compounding - is a real, profitable business.



8.3 Beyond month 12: the fleet illustration

Mechanically extending the base curve: box #2 arrives at M13 (171 customers), #3 at M16, and by M24 the fleet is ~20 boxes serving ~3,100 customers at ~€67,000 MRR - each box self-funding from the €3.3k/month surplus of the previous ones. This is an illustration of the capex-light scaling property, not a forecast: 30% monthly compounding will not survive two years of contact with reality, and the operator becomes the bottleneck long before box #20 (see R8). The honest reading: **the marginal economics of growth are excellent, and growth is never gated on capital.**

9. Sensitivity analysis

Each subsection perturbs one assumption with everything else held at base. Tables were recomputed from a programmatic replica of the model of record; base-case columns reproduce the sheet exactly.

9.1 Churn

Monthly churn	3%	5% (base)	8%	10%
LTV (€)	674	405	253	202
Max sane CAC (€)	225	135	84	67
M12 paying customers	138	130	122	115
M12 monthly profit (€)	2,789	2,623	2,458	2,313

Churn barely dents year-one P&L but demolishes the acquisition budget: at 8% the max CAC falls under €85, ruling out most paid channels. Retention is the long-game lever; watch cohort curves from the first ten customers.

9.2 Average snapshot size (the most dangerous number)

Avg snapshot	100 MB	130 MB (base)	200 MB	300 MB	500 MB
Parked sandbox capacity	4,628	3,560	2,314	1,542	925
Customer ceiling / box	212	163	106	70	42
Revenue ceiling / box (€)	4,579	3,521	2,290	1,512	907

A PyTorch-sized dependency tree averaged into the fleet would halve the business per box. Mitigations (R2): per-plan rootfs quotas, deduplicated base layers, cold-snapshot eviction to object storage, and a monthly p95 review. This is a policy problem, not a hardware problem - enforce it in the builder.

9.3 Resident-concurrency drift (the scale-to-zero bet)

Actual load vs plan	x1.0 (base)	x1.5	x2.0	x3.0
Resident VMs / customer	0.55	0.83	1.11	1.66
Concurrency-bound customers	217	144	108	72
Binding ceiling / box	163 (disk)	144 (RAM)	108 (RAM)	72 (RAM)
Contribution margin	93.7%	92.7%	91.8%	89.9%

If customers idle half as much as assumed, margins barely notice but the ceiling drops by a third - capacity, not profitability, absorbs the error. Sheet logic shown; under the demand-weighted blend of section 6.2 the flip to RAM-bound happens even at x1.0.

9.4 Customer mix

Mix (Hobby/Pro/Scale)	70/25/5	55/40/5 (base)	40/50/10
Blended ARPU (€)	18.60	21.60	28.10
Contribution (€)	17.37	20.23	26.30
Customer ceiling / box	189	163	124
Profit at ceiling (€/mo)	3,293	3,306	3,282

Striking result: profit per full box is mix-invariant at ~€3.3k - plan quotas scale with price, so the box converts its RAM and disk into the same money regardless of logo count. Prefer the Pro-heavy world anyway: the same revenue from 124 customers instead of 189 means fewer support threads

and fewer churn events.

9.5 Stripe: non-EU vs EU entity

	2.9% + €0.30 (base)	1.5% + €0.25 (EU entity)
Stripe cost / customer (€)	0.926	0.574
Contribution margin	€20.23 (93.7%)	€20.59 (95.3%)
M12 monthly profit (€)	2,623	2,669
Profit at ceiling (€/box)	3,306	3,363

Worth €0.35 per customer per month, forever - roughly €57/month per full box. Do it when incorporating anyway (section 12); it is not urgent enough to gate launch.

9.6 Hobby price: €9 vs €12

	Hobby €9 (base)	Hobby €12
Blended ARPU (€)	21.60	23.25
M12 MRR (€)	2,808	3,023
M12 monthly profit (€)	2,623	2,832
Revenue at ceiling (€)	3,521	3,790

+€209/month at M12 if conversion is unharmed. Compromise: launch at €9 to buy the first hundred logos and testimonials, move new signups to €12 at Gate B with grandfathering - the sensitivity says the model does not need the raise to work.

9.7 Free-tier ratio

Free users per paying	2	4 (base)	8
Cost to serve / customer (€)	0.433	0.441	0.458
Customer ceiling / box	180	163	138

Doubling free load costs 1.7 euro-cents per customer but 25 customers of ceiling: the free tier is paid for in disk slots. Policy answer: idle free sandboxes auto-delete after ~14 days of zero traffic (one click to redeploy), which reclaims most slots without hurting the funnel.

9.8 Founder salary (what ramen mode is worth)

Monthly draw (€)	0 (base)	500	1,000	2,000
Break-even customers	4	28	53	103
Base scenario covers it from	M2	M7	M9	M12
As % of one box's ceiling	2.5%	17%	33%	63%

A €2,000 draw needs 103 customers - still well inside one box. The zero-salary figures elsewhere in this paper are honest only if runway comes from elsewhere for roughly the first year; this table is the bridge to a livable income.

9.9 Server price

EX44 quote	€39.44 + €39 setup	€44 + €39 (base)	€50 + €99 setup
Fixed cost / month (€)	59.69	64.25	75.25
COGS per GB-h (€)	0.00172	0.00185	0.00217
Break-even customers	3	4	4

The business is indifferent across any plausible Hetzner quote. Grab a €0-setup promo if one appears; do not wait for one.

9.10 What the sensitivity grid says as a whole

The model is margin-robust and capacity-fragile. No plausible perturbation of prices, fees, or server cost threatens profitability - break-even never leaves single digits without a salary line. Every real threat is physical: snapshot bytes, resident RAM, free-tier disk slots. Operational conclusion: the engineering priorities that protect the P&L are quota enforcement in the builder, idle eviction, and telemetry on the three C-grade assumptions - not feature work.

10. Competitive landscape (July 2026)

10.1 Unit-price anchors

Provider	Pricing model	Effective rate	vs our COGS
Us - COGS	fully loaded production cost	€0.00185 / GB-h	1.0x
Us - overage meter	metered beyond plan quota	€0.01 / GB-h	5.4x
Fly.io	RAM adder \$5/GB/30d (+CPU presets)	\$0.007-0.012 / GB-h	~4-6x
Railway	\$10 / GB RAM / month, per-second	\$0.0137 / GB-h	~7x
AWS Lambda (x86, tier 1)	\$0.0000166667 / GB-s + \$0.20/M req	\$0.060 / GB-h	~32x
Cloudflare Workers	\$5/mo incl 10M req + 30M CPU-ms	CPU-time, not RAM-time	n/a
Render	fixed instances from ~\$7/mo	always-on, no scale-to-zero	n/a

Native currencies; EUR/USD near parity does not change the ordering. Lambda's first pricing tier shown; ARM is ~20% cheaper. Cloudflare bills V8-isolate CPU-milliseconds - unbeatable for I/O-bound edge code, but not a general Linux runtime.

10.2 Positioning

	Us	Fly.io	Railway	Lambda	CF Workers
Isolation	Firecracker microVM	Firecracker microVM	container	Firecracker microVM	V8 isolate
Scale to zero	yes (snapshot)	yes	partial	yes	yes
Full Linux runtime	yes	yes	yes	restricted	no
Global edge	no (1-2 DCs)	yes	regions	yes	yes (300+)
Egress fees	none (1 Gbit flat)	yes	yes	yes	mostly none
EU-metal residency	yes (DE/FI)	partial	partial	regional	network-wide
Pricing anxiety	flat plans	usage	usage	usage	usage

Where we win: many small, idle-heavy, full-runtime apps (any language, real Linux, background processes) parked cheaply with instant wake; developers who want a flat bill; EU data residency on physical metal; zero egress anxiety. **Where we lose, and should not fight:** global edge latency, five-nines availability promises, compliance checklists, CPU-pegged or egress-heavy workloads, and anything needing a managed-services catalogue.

10.3 An honest note on moats

There is no technical moat: Firecracker is open source and the architecture is reproducible by any competent engineer reading this document. The defensible assets are the cost structure (metal at 1/30th of hyperscaler RAM-time, no payroll), the niche focus, accumulated operational scar tissue, and - if cultivated - a community and reputation for support quality that big platforms structurally cannot match at this price point. That is a real small-firm moat, and it is enough for the ambition stated here; it would not survive a funded competitor targeting the same niche with the same architecture. Speed to a loyal first hundred customers matters.

11. Risk register and mitigations

Likelihood x impact grading; triggers are the metrics that page the operator. The first three risks are structural and were flagged in review; the rest emerge from operating the architecture.

#	Risk	L x I	Mitigation	Trigger metric
R1	Build-phase OOM: a heavy build (npm, pip wheels) starves production microVMs on the shared box	H x H	Global build semaphore (max 2), 2 GB cgroup per build, 100 MB deploy cap on Free/Hobby, kill-and-requeue on timeout; remote builder VM if queues hurt	Host memory pressure during builds; build queue p95
R2	Snapshot-size drift: real apps balloon past the 130 MB average and halve the per-box ceiling	H x H	Per-plan rootfs quotas enforced in the builder, dedup of shared base layers, cold-snapshot eviction to object storage, monthly p95 review	Fleet avg and p95 snapshot MB vs 130 plan
R3	Single point of failure: one motherboard is the whole company until a second box exists	M x Crit	Gate A (~20 paying, MRR €430+): second EX44 in a different Hetzner DC, active-passive - streaming Postgres replica, hourly snapshot sync, DNS failover; RTO under 30 min. Interim: tested restore runbook from Storage Box + status page	Time-to-restore drill result; replication lag
R4	Noisy neighbours and cryptominers attracted by cheap compute	H x M	Per-VM CPU quotas, anomaly alerts on sustained 100% CPU, AUP banning mining, SMTP/25 and risky egress blocked, card verification before meaningful CPU	Sustained-CPU tenant count; abuse reports
R5	microVM escape or platform breach (untrusted code on shared metal)	L x Crit	Firecracker + jailer, seccomp filters, non-root VMM, minimal guest kernels, fast patch policy on Firecracker/kernel CVEs; blast radius = one box; honest disclosure plan	CVE feed; unexpected host syscalls (auditd)
R6	Hetzner dependency: abuse-desk null-route or account action takes everything down	L-M x H	Outbound abuse prevention (R4), responsive abuse-contact handling, second box in second DC/location, backups restorable to any provider	Abuse tickets; provider status
R7	Free-tier signup abuse (disposable accounts farming sandboxes)	H x L-M	Email + GitHub-age or card check, per-IP caps, 1 sandbox limit, idle eviction after ~14 days zero traffic	Signups/IP; free-tier disk share
R8	Support load scales with logos, not revenue - solo operator saturates	M x M	Docs-first culture, self-serve everything, status page, community channel; observed metric budget ~1 support-hour per 15 customers/mo; Pro-heavy mix preferred (9.4)	Tickets per 100 customers; founder hours/wk
R9	Compliance: EU VAT on digital services from customer #1, GDPR as a data processor	M x M	Incorporate early; Stripe Tax or quarterly OSS filing; DPA template + subprocessor list (Hetzner); data stays in EU by default	Revenue by country; DPA requests
R10	DDoS or bandwidth saturation of the shared 1 Gbit uplink	M x M	Hetzner's included DDoS protection, per-tenant rate limits at the gateway, free tier behind Cloudflare proxy if targeted	Uplink p95; gateway 429 rate

The pattern across the register: every high-likelihood risk is mitigated by *limits enforced in software*, and every high-impact risk is mitigated by *a second box and tested restores*. Both are cheap. The expensive failure mode is skipping them while chasing features.

12. Strategic roadmap and decision gates

Months 0-1: foundation

Incorporate (an EU entity cuts Stripe to ~1.5% + €0.25 and simplifies VAT OSS). Capture the EX44 - current list ~€39.44; take a €0-setup promo if available at ordering time. Build the minimum loop that is genuinely the product: deploy → build → snapshot → wake-on-request → meter → invoice. Ship with the day-one limits that the sensitivity grid says protect the business: 100 MB deploy cap, build semaphore, per-plan rootfs quotas, per-VM CPU shares, free-tier idle eviction. Instrument the three C-grade assumptions from day one.

Months 2-3: first ten customers

Hand-hold the first ten paying customers personally; their workloads calibrate the fragile numbers (snapshot p95, resident concurrency, build memory). Break-even (4 customers) should fall in this window. Publish the pricing page with the €9 Hobby anchor and honest limits - the cheap-and-transparent position is the marketing.

Gate A - second box (availability gate). Trigger: ~20 paying customers or MRR €430+ (covers box #2 roughly eightfold at contribution margin; base case hits this around M6). Action: second EX44 in a different Hetzner location, active-passive failover, restore drill documented. This gate is about surviving a motherboard, not about capacity.

Months 4-6: self-serve

Docs-first onboarding, status page, automated invoicing and dunning, community channel. Free-tier funnel opened wider once idle-eviction is proven. Target: 20-25 paying customers by M6 (base case: 23), founder time shifting from build to write - content is the €0-CAC channel, and the LTV:CAC math (max €135) says paid channels are marginal at this ARPU.

Gate B - pricing gate. Trigger: ~100 paying customers. Action: new signups move to Hobby €12 (grandfather existing), revisit mix - section 9.4 says nudging customers toward Pro beats adding logos. Consider a dedicated build VM if queue p95 exceeds a minute.

Months 7-12: compounding

Aim for the base curve: 130 paying by M12, €2,800 MRR, 60% of box #1's resident capacity (83% demand-weighted - watch section 6.2's stricter gauge). Begin a founder draw once MRR sustainably clears €1,500 (€1,000 draw breaks even at 53 customers). Keep the sheet as the source of truth: monthly, replace assumptions with telemetry, re-derive the ceiling, and re-date Gate C.

Gate C - capacity gate. Trigger: ~140 paying customers on one box (85% of the practical ~156-customer RAM ceiling). Action: box #3 or snapshot eviction (cuts per-customer disk 30-40%), whichever telemetry says binds first.

Month 13+: choose the ambition

Two legitimate endgames. **Lifestyle:** cap at 2-3 boxes, ~€7-10k/month profit, one operator, deliberately boring. **Growth:** reinvest the surplus into DX and content, ride the fleet illustration of section 8.3, hire support at ~300 customers. The unit economics fund either; the model does not force the choice before it is real. What it does force: never let the three C-grade numbers drift unwatched, because they are the difference between the two futures.

13. Appendix

A. Model formulas (as implemented in the sheet)

Quantity	Formula	Base value
Fixed monthly cost	rent + setup/12 + backup + domain + object + monitoring	€64.25
Tenant RAM budget	64 - 2 (host) - 6 (control plane)	56 GB
Sellable GB-RAM-hours / mo	56 x 0.85 x 730	34,748
COGS per GB-RAM-hour	64.25 / 34,748	€0.0018490
Avg resident VM (sheet blend)	$0.55 \times 256 + 0.40 \times 512 + 0.05 \times 1024 + 8$ overhead	404.8 MB
Safe resident VMs	$\min(56 \times 1024 \times 0.85 / 404.8 ; 20 \times 8)$	120 (RAM)
Concurrency / customer	$0.55 \times 0.15 + 0.40 \times 0.60 + 0.05 \times 3.0 + 4 \times 0.02$	0.5525
Parked snapshot capacity	$(512 - 60) \times 1024 / 130$	3,560
Deployed / customer	$0.55 \times 5 + 0.40 \times 25 + 0.05 \times 100 + 4 \times 1$	21.75
Customer ceiling	$\min(120 / 0.5525 ; 3560 / 21.75)$	163 (disk)
Blended ARPU	$0.55 \times 9 + 0.40 \times 29 + 0.05 \times 99 + 10 \times 0.01$ overage	€21.60
Cost to serve / customer	plan-weighted cost + 4 x free cost + overage COGS	€0.4411
Contribution / customer	$21.60 - 0.4411 - (21.60 \times 2.9\% + 0.30)$	€20.2325
LTV / max CAC	contribution x (1/churn) ; LTV / 3	€404.65 / €134.88
Break-even customers	$\text{ceil}(64.25 / 20.2325)$	4

B. Sheet-to-section mapping

Workbook tab	This document
Assumptions	Section 5 (register), 2.2 (server), 9 (perturbations)
Pricing	Sections 3, 7.2-7.4
Capacity	Section 6 (including the integrity note, 6.2)
Costs	Section 7.1
Projection	Section 8
Unit economics	Sections 7.4-7.5, 9.1

C. External sources (accessed July 2026)

Hetzner EX44 specification and pricing: hetzner.com/dedicated-rootserver/ex44 (list tracking via whotop.com; Hetzner price-adjustment notice of 15 June 2026 at docs.hetzner.com). AWS Lambda pricing: aws.amazon.com/lambda/pricing. Fly.io resource pricing: fly.io/docs/about/pricing. Railway plans: docs.railway.com/pricing/plans. Cloudflare Workers pricing: developers.cloudflare.com/workers/platform/pricing. Render pricing: render.com/pricing. Firecracker characteristics (boot time, overhead, creation rate): firecracker-microvm.github.io and fly.io/learn/firecracker-vm. Stripe EU and non-EU card rates: stripe.com/pricing. All third-party prices are as published at access time and should be re-verified before external use.

End of document. Keep it honest: whenever the spreadsheet changes, this paper is stale until regenerated.